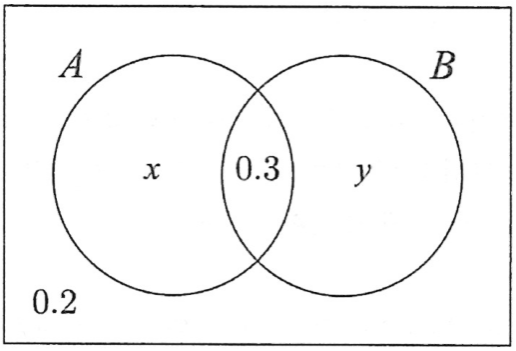
Probability: Conditional Probability

Questions 1–31

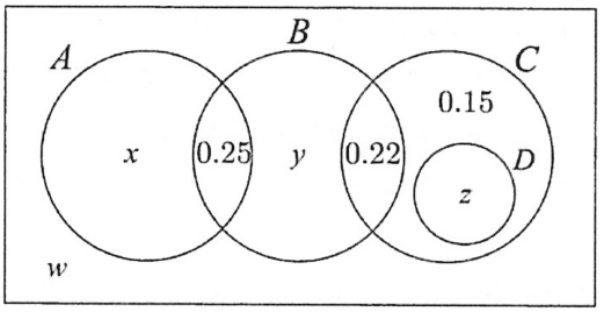
Question 1. It is given that A and B are two events such that $\mathrm{P}(A) = \frac{1}{8}$, $\mathrm{P}(B) = \frac{1}{2}$ and $\mathrm{P}(A \cap B) = \frac{1}{10}$.
Find $\mathrm{P}(A \mid B)$

Question 2. It is given that A and B are two events such that $\mathrm{P}(A) = \frac{1}{4}$, $\mathrm{P}(B) = \frac{2}{5}$ and $\mathrm{P}(A \cup B) = \frac{1}{2}$.
Find $\mathrm{P}(B \mid A)$

Question 3. Events A and B are shown in the Venn diagram below. It is given that $\mathrm{P}(A \mid B) = 0.5$. Find the value of x and the value of y



Question 4. Events A, B, C and D are shown in the Venn diagram below.



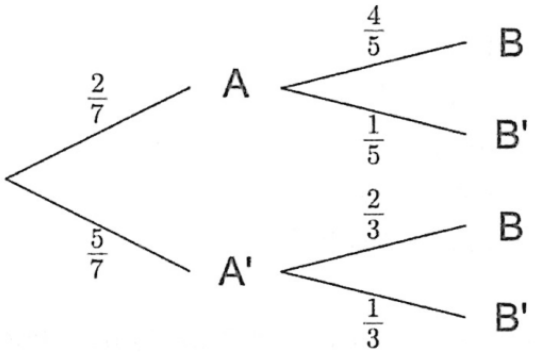
a) Given $\mathrm{P}(C) = 0.4$, find z
b) Given B and C are independent, find y
c) Given $\mathrm{P}(B \mid A) = 0.625$, find x
d) Find w

Question 5. At a school, students must choose their GCSE options in Year 9. As part of their choices, they must choose to study History, or Geography, or both, or neither. The probability of a student choosing History is $\frac{1}{4}$ and the probability of a student choosing Geography is $\frac{2}{5}$. The probability of a student choosing History, given that they have chosen Geography, is $\frac{1}{3}$.
Find the probability that a student picked at random chose neither History or Geography.

Question 6. It is given that A and B are two events such that $\mathrm{P}(A \mid B) = 0.3$, $\mathrm{P}(B \mid A) = 0.4$ and $\mathrm{P}(A' \cap B') = 0.1$.
Find $\mathrm{P}(A \cap B)$

Question 7. Using the following probability tree, find:

a) $\mathrm{P}(A \mid B)$
b) $\mathrm{P}(B \mid A)$
c) $\mathrm{P}(A \mid B')$
d) $\mathrm{P}(B' \mid A)$
e) $\mathrm{P}(A' \mid B')$



Question 8. In a pack of ten playing cards, 8 are blue and 2 are red. One card is randomly chosen from the pack and is not replaced. Another card is then randomly chosen from the pack and not replaced. Find the probability that the first card was blue, given that the second card was red.

Question 9. The Maxi box of chocolates has 60 chocolates with nine different varieties.

	Milk	Dark	White	Total
Plain	16	6	6	28
Caramel	7	4	4	15
Hazelnut	7	5	5	17
Total	30	15	15	60

Find the probability that a randomly selected chocolate

a) is a plain milk chocolate;
b) is a plain chocolate;
c) is a white chocolate, given that it is a plain chocolate;
d) is a hazelnut chocolate, given that it is a white chocolate.
e) Write down two events from the table that are mutually exclusive.

Question 10. It is given that A and B are two events such that $\mathrm{P}(A \cap B) = 0.25$, $\mathrm{P}(A' \cap B) = 0.3$, and $\mathrm{P}(A) = \mathrm{P}(B')$.

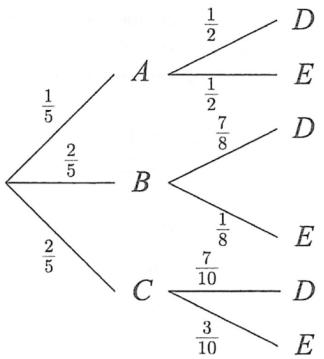
a) Find $\mathrm{P}(B)$
b) Find $\mathrm{P}(A \mid B)$
c) Determine if A and B are independent events.

Question 11. It is given that $\mathrm{P}(A) = \frac{4}{5}$, $\mathrm{P}(B) = \frac{2}{3}$, and $\mathrm{P}(A \cap B) = \frac{11}{20}$

a) Find $\mathrm{P}(A \mid B)$
b) Find $\mathrm{P}(B \mid A)$
c) Find $\mathrm{P}(A \mid B')$
d) Find $\mathrm{P}(B' \mid A')$
e) Find $\mathrm{P}([A \cup B]')$

Question 12. Using the following probability tree, find:

a) $\mathrm{P}(A \mid D)$
b) $\mathrm{P}(B \mid E)$
c) $\mathrm{P}(C \mid D)$
d) $\mathrm{P}(A' \mid E)$



Question 13. A large pottery makes bowls from two types of clay: terracotta and porcelain. The proportion that are made with terracotta is 73% and the proportion that are made with porcelain is 27%. During the firing process, the probability that a terracotta bowl cracks is 18% and the probability that a porcelain bowl cracks is 32%.
One bowl is chosen at random.

a) Find the probability that it is terracotta, given that it is cracked.
b) Find the probability that it is porcelain, given that it is not cracked.

Question 14. It is given that A and B are two events such that $\mathrm{P}(A) = \frac{7}{10}$, $\mathrm{P}(B) = \frac{2}{3}$ and $\mathrm{P}(A \cap B) = \frac{3}{5}$

a) Find $\mathrm{P}(A \mid B)$
b) Use your answer to a) to determine whether the events A and B are independent.

Question 15. It is given that A and B are two events such that $\mathrm{P}(A) = \frac{11}{15}$, $\mathrm{P}(B') = \frac{5}{11}$ and $\mathrm{P}(A' \cap B) = \frac{8}{55}$

a) Find $\mathrm{P}(A' \mid B)$

b) Use your answer to a) to determine whether the events A and B are independent.

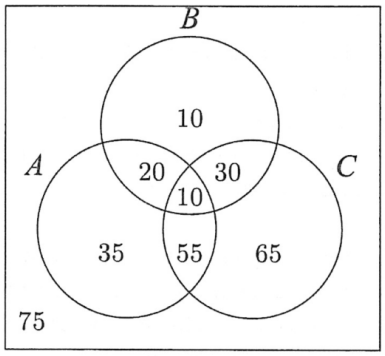
Question 16. It is given that A and B are two events such that $\mathrm{P}([A \cup B]') = \frac{1}{25}$, $\mathrm{P}(A' \cap B) = \frac{2}{5}$ and $\mathrm{P}(B) = \frac{2}{3}$. Find $\mathrm{P}(A \mid [A \cap B]')$

Question 17. At an auction house, two sets of cutlery, one set that is silver and one set that is silver-plated, are accidentally mixed into the same box. The silver set has four forks, four knives and four spoons. The silver-plated set has six forks, six knives and six spoons. A piece of cutlery is selected at random from the box. Find the probability that it is silver, given that it is a fork or a knife.

Question 18. Tim made 2 cups of tea and 3 cups of coffee in five identical mugs and forgot which is which. He decides to randomly choose a mug and smell whether it is coffee or not, which he can identify with 100% accuracy. He will continue to randomly select mugs until he is sure he has identified 2 mugs of tea or 3 mugs of coffee.

a) Draw a probability tree to display this information.
b) Find the probability that Tim must check four mugs.
c) It is given that Tim only checks three mugs. Find the probability that the last mug he checks is tea.
d) Given that the second mug he checks is coffee, find the probability that the first mug was tea.

Question 19. Events A, B, C and D are shown in the Venn diagram below.



Find:

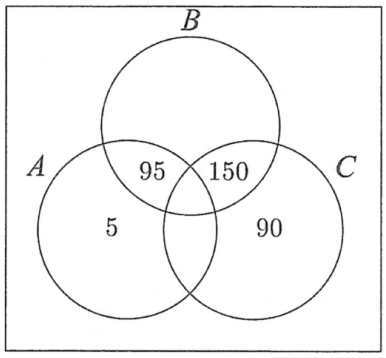
a) $\mathrm{P}(A \cap B)$
b) $\mathrm{P}(A \mid B)$
c) $\mathrm{P}(B \mid A)$
d) $\mathrm{P}(C \mid B)$
e) $\mathrm{P}(B \mid C)$
f) $\mathrm{P}(A \mid [B \cup C])$
g) $\mathrm{P}(B \mid [A \cup C])$
h) $\mathrm{P}(B \mid [A \cap C])$
i) $\mathrm{P}([B \cup C] \mid A)$
j) $\mathrm{P}([B \cap C] \mid A)$
k) $\mathrm{P}([A \cap B] \mid [A \cup B])$

Question 20. In a sixth form college, the numbers of students studying Psychology, Physics and Chemistry in Year 12 and 13 are shown in the table below.

	Year12	Year13
Psychology	250	200
Physics	100	80
Chemistry	60	40

All students at the college take at least one of Psychology, Physics, Chemistry and Maths. It is given that 58% of Psychology students, in both years, study A-Level Maths. 95% of Physics students, in both years, study A-Level Maths. 85% of Chemistry students, in both years, study A-Level Maths.
Event A is that the student studies Physics.
Event B is that the student studies Maths.
Event C is that the student is in Year 13.

a) Use this information to complete the Venn diagram below.



Question 20 (continued).

b) Find $\mathrm{P}(A \cap B \cap C)$

c) Find $\mathrm{P}(A' \cap B \cap C)$

d) Find $\mathrm{P}([A \cup B] \mid [B \cup C])$

e) Find $\mathrm{P}([A \cap B] \mid [B \cap C])$

Question 21. It is given that event A is mutually exclusive with both of the events B and C. It is also given that events B and C are independent. It is given that $\mathrm{P}(A) = \frac{1}{12}$, $\mathrm{P}(B) = \frac{1}{14}$ and $\mathrm{P}(C) = \frac{1}{16}$.
Find $\mathrm{P}([A \cup B] \mid [B \cup C])$

Question 22. It is given that A and B are two events such that $\mathrm{P}([A \cap B']') = \frac{8}{9}$, $\mathrm{P}(A' \cap B') = \frac{1}{10}$ and $\mathrm{P}(A) = \frac{1}{5}$.
Find $\mathrm{P}([A \cup B] \mid [A \cup B'])$

Question 23. It is given that A, B and C are three events such that $\mathrm{P}(A \cap B \cap C) = \frac{1}{13}$, $\mathrm{P}(A \cap B) = \frac{18}{65}$, $\mathrm{P}(A \cap C) = \frac{3}{13}$ and $\mathrm{P}(B \cap C) = \frac{11}{130}$.
It is also given that $\mathrm{P}(B \mid A) = \frac{36}{71}$, $\mathrm{P}(A \mid B) = \frac{36}{41}$ and

$$4\mathrm{P}(B \cap [A \cup C]') = \mathrm{P}(C \cap [A \cup B]')$$

Draw a fully labelled Venn diagram for the events A, B and C.

Question 24. It is given that A and B are two events such that $\mathrm{P}(A \mid B) = \frac{4}{11}$ and $\mathrm{P}(B \mid A) = \frac{40}{49}$. It is also given that $\mathrm{P}([A \cup B]') = \frac{3}{20}$. Draw a fully labelled Venn diagram for the events A and B.

Question 25. Ellen is throwing darts at a darts board. The probability that she hits the bullseye with a single dart at a horizontal distance of 200 cm is $\frac{1}{6}$. Every 10 cm that she steps away from the darts board, the probability of her hitting the bullseye reduces by 15%. She plays a game where only if she hits the bullseye can she move 10 cm further away from the darts board to take the next shot. Given that she reaches 230 cm, find the probability that she hits the bullseye with her next shot.

Question 26. On Sunday, Sarah gets to choose between three different activities: hiking, watching a movie, or attending a music concert. The probability of her choosing hiking is 0.35, the probability of choosing a movie is 0.4, and the probability of attending a concert is 0.25. If she goes hiking, the probability of her feeling tired the next day is 0.8. If she watches a movie, the probability of her feeling tired the next day is 0.2. If she attends the music concert, the probability of her feeling tired the next day is 0.6. On Monday, Sarah felt tired. Determine the probability that she chose to go hiking for her on Sunday.

Question 27. On a mysterious island, 7.9% of the population have developed a unique psychic ability known as "Mindlink". 89.6% of the community has participated in a special meditation practice associated with developing Mindlink, while 9.2% of the community has neither engaged in the meditation practice nor possess the psychic ability. Emily has participated in the meditation practice. Find the probability that she will develop the Mindlink ability.

Question 28. Olena's pencil case contains several gel pens and felt tip pens of varying colours, as shown in the table below.

	Red	Green	Blue
Gel	3	1	1
Felt-tip	4	2	2

Olena takes four pens out of the pencil case at random.

a) Find the probability that all four colours are red.
b) Find the probability that exactly three gel pens are picked.
c) Find the probability that exactly two felt-tip pens are selected, given that all three blue pens are selected.

Question 29. In a television game show, a contestant is asked a series of multiple-choice questions, where each question has four possible answers. Valerie is playing the game and the probability that she knows the answer to the first question is $\frac{3}{8}$. The more questions she answers, the harder they become, so that the probability that she knows the answer is 10% less than the probability before. If she does not know the answer, she guesses. Given that Valerie reaches Question 4, find the probability that she guessed the answer to Question 3.

Question 30. Priya has a bag that contains 28 blue marbles and n red marbles. There are fewer red marbles than blue marbles in the bag. Priya picks a marble at random from the bag and does not replace it. She then randomly selects another marble from the bag. The probability that the two marbles are the same colour is 0.5. Find the probability that the 1st marble is blue, given that the 2nd marble is red.

Question 31. Issy has a box of assorted chocolates that contains 63 milk chocolates and n dark chocolates. There are more dark chocolates than milk chocolates in the box. Issy picks a chocolate at random from the box and eats it. She then randomly selects another chocolate from the box. The probability that the two chocolates are of different types is 0.35. Find the probability that 1st chocolate is milk, given that the 2nd chocolate is milk.